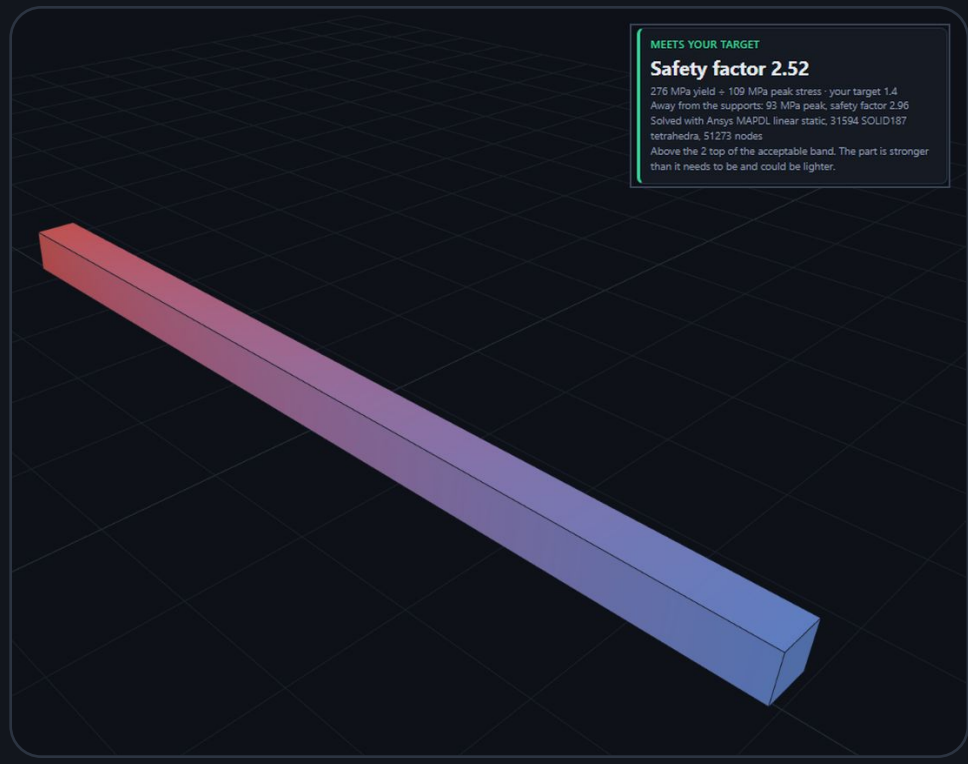
LIFE IN SPACE · TEAM 97070 · GEAUX HACK 2026

SimForge

Will this part hold on the Moon? Ask in one sentence.

Import a STEP file, name the destination, and describe the test. SimForge sets up the Ansys analysis and returns a safety factor in minutes.

Arav Gandhi · Kevin Guo · Braxton Anderson · Amr Mohieldin · Jeremy Lin



A habitat only works if its structure holds

Gravity changes

Moon 1.62 m/s², Mars 3.73 m/s². The same 500 kg rack loads its mount about 2.3 times harder on Mars.

Pressure pushes out

Every pressurized wall has cabin air on one side and almost nothing on the other.

Checks are slow

Setting up a finite element check by hand takes expertise. We turn it into a conversation.

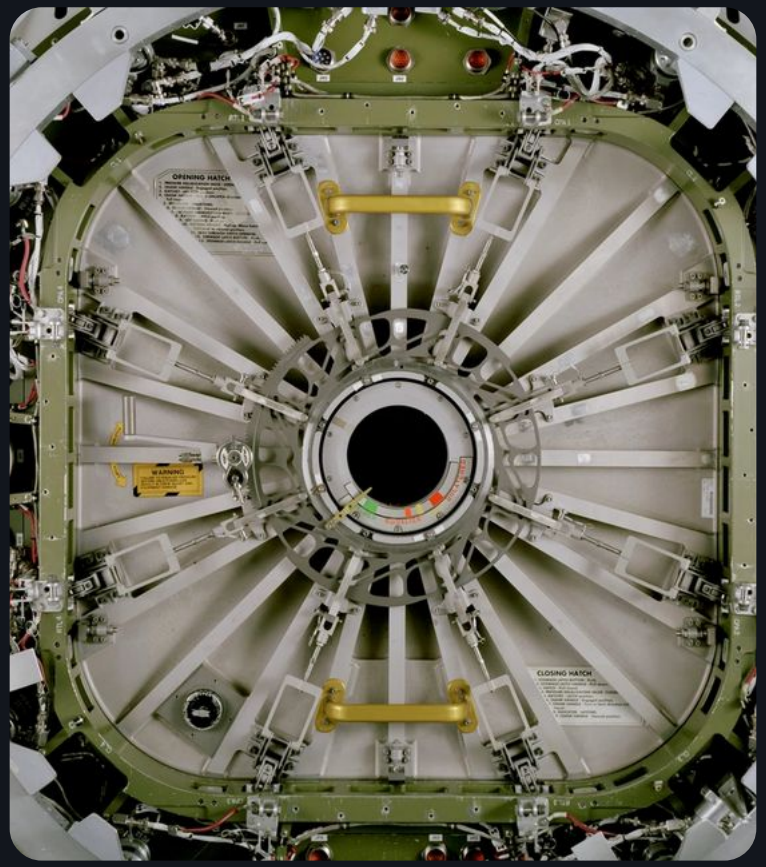


Image: NASA, JSC2001-02112. ISS common hatch in the Space Station Systems Trainer.

Built for Life in Space

Help humans sustain life in extreme off-Earth environments

Hostile environments

The Moon, Mars, Mercury, Venus, Pluto and free space, each with gravity and outside pressure from NASA fact sheets.

Habitat metrics

Cabin pressures from NASA documents: 14.7 psia ISS-type, 8.0 psia lunar habitat, 7.6 psia Mars habitat.

Crew consequences

Each result says what a failure would mean for the people inside, such as a hatch letting cabin pressure fall.

Any other world works too: type its name and give its surface gravity and outside pressure.

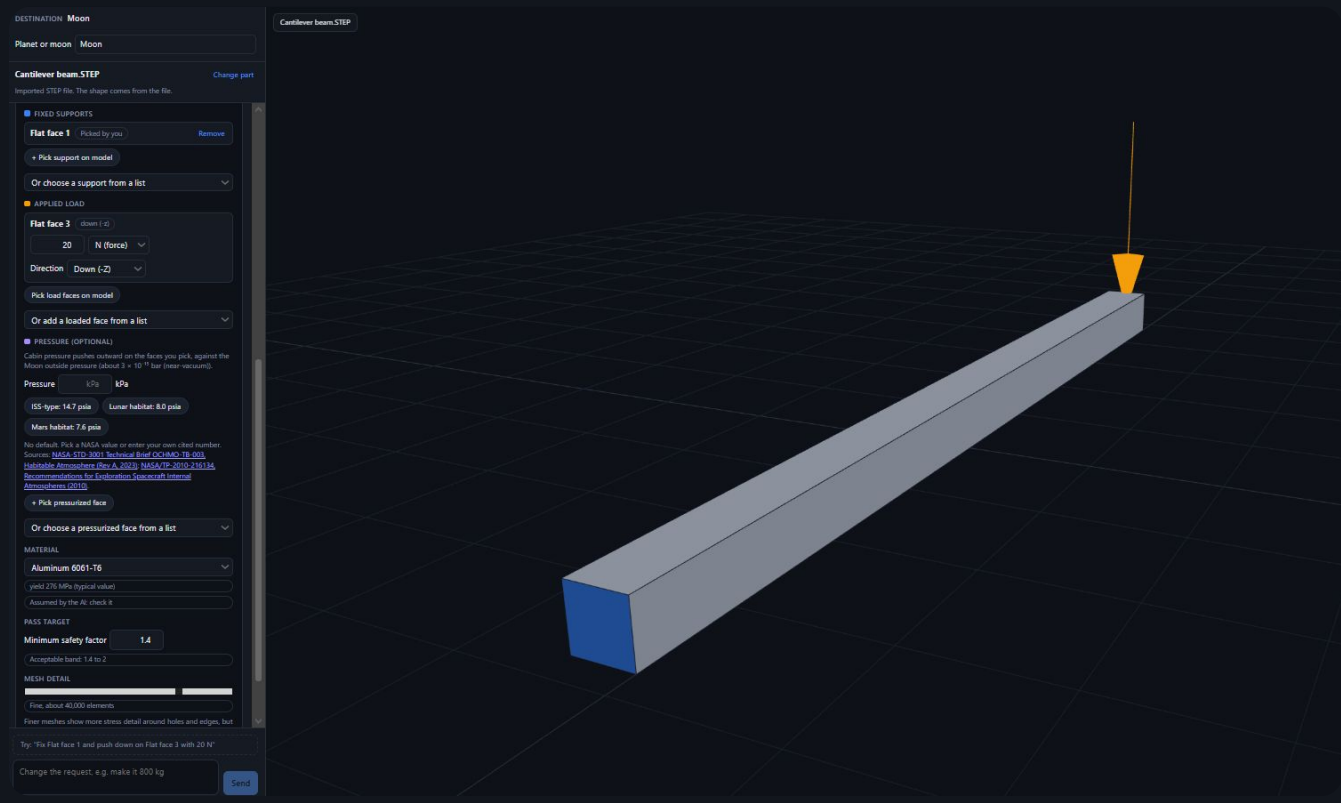
How it works

1 Describe
Import a STEP file, type a destination, and write the test in plain English.

2 Propose
Google Gemini picks the fixed faces, the load and the material, and lights them up in 3D.

3 Confirm
You check the faces on the model, change any by clicking it, and approve.

4 Analyze
Ansys cuts the part into tiny pieces and solves the stress in each one.



Tech stack: what happens under the hood

Browser
React, TypeScript and three.js draw the 3D model

→

Backend
Python, FastAPI and Open CASCADE read the STEP file

→

Gemini
Google Gemini proposes faces, load and material from your sentence

→

Mesh
gmsh cuts the part into tiny four-sided blocks

→

Ansys
Ansys MAPDL, driven by PyMAPDL, solves the stress in every block

A dial for detail
Coarse is about 5,000 blocks and runs fast. Finest is about 65,000 and shows more detail. You choose.

Self-repairing meshes
Curved parts can produce a block Ansys rejects. We straighten its edges automatically so the run continues.

Backups at every step
If Gemini is busy, a lighter model answers. If both fail, plain rules do, and the chat says so.

Ansys Student allows one session at a time. If an old session blocks it, the backend ends the stale one and restarts Ansys by itself.

Safety factor decides the verdict

Yield strength divided by peak stress

Below 1

Fail

The part yields.

1 to 1.4

Below target

Holds, with too little margin.

1.4 to 2

Acceptable

Our target band.

Above 2

Heavier than needed

Passes with margin to spare.

The 1.4 to 2 band is our own requirement. For comparison, NASA-STD-5001B asks for 1.4 on ultimate strength and 1.25 on yield for protoflight metallic hardware. We apply 1.4 to yield, which is stricter.

LIVE

Demo

1. Import the beam STEP file
2. Type the destination: the Moon
3. Describe the test in plain English
4. Confirm the highlighted faces
5. Run Ansys, read the safety factor

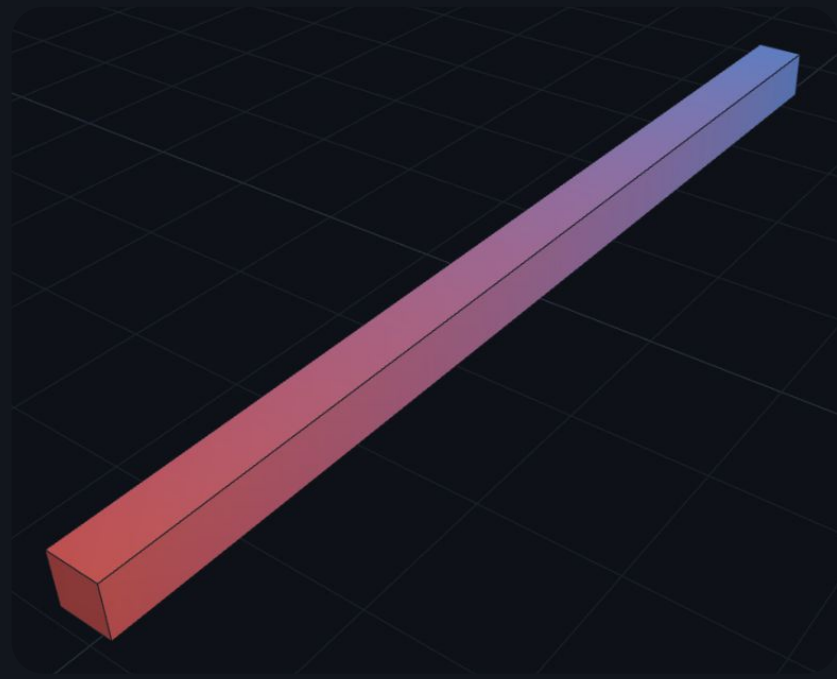
We checked the answer against a hand calculation

Tip deflection, under 1% apart

1.85 mm	1.86 mm
Ansys	Beam theory

Peak stress, about 3% apart

93 MPa	96 MPa
Ansys	Beam theory



6061-T6 aluminum beam, 100 x 5 x 5 mm, 20 N at the free end, 31,594 mesh blocks. Stress is read away from the fixed end, where a rigid support overstates it (109 MPa at the edge).

The AI never decides the verdict

The AI picks

Faces, a load direction and a starting material (flagged "check it"), plus the numbers you wrote. Gemini 3.8 Flash, with 3.5 Flash-Lite as backup.

Ansys solves

The stress and the safety factor come from the solver.

Sources supply

Gravity and outside pressure from NASA. Material values from datasheets.



Sources: NASA-STD-5001B · NASA-STD-3001 Technical Brief · NASA/TP-2010-216134 · NASA planetary fact sheets

Challenges we faced

Ansys Student limits

One session at a time and 128,000 nodes. A stale session crashed the solve, so we detect it, end it and retry automatically.

Bad blocks on curves

Ansys rejected a few blocks on the nut and kettlebell. We turned on gmsh optimization and wrote a step that straightens them.

Rigid supports lie

The peak at a fixed face grows as the mesh gets finer. The app reports the peak away from the supports too, and says which is which.

No made-up numbers

We removed an invented default load and confidence score. A load with no number stays empty until you type one.

Also: Gemini can be busy, so there is a timeout, a backup model and plain rules. The demo never hangs.

Built to scale

Any part

Any STEP file. The mesh dial runs from about 5,000 to 65,000 blocks, so small parts stay fast and detailed ones get more.

Any world

Type a name and give its gravity and outside pressure. Six worlds and free space are built in from NASA fact sheets.

Any material

Two are built in today: aluminum 6061-T6 and A36 steel. Each is one entry in a data file, so adding more is simple.

What comes next

Today one analysis runs at a time because Ansys Student allows one session. A full Ansys license, or a queue of solver machines, lifts that limit. We would then add contact, bolts, thermal loads and buckling.

Who it helps

Student and hobby teams

Check a design before building it, without learning simulation setup.

Small aerospace startups

Get a first stress answer in minutes, before booking an analyst.

Engineers new to simulation

Review the setup the copilot proposes instead of building it from scratch.



Same L bracket, same 30 kg load, same mesh. The Moon gives a safety factor of 3.19. On Mars the stress is 2.3 times higher, so the same part drops to 1.38, below our target.

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Thank you

Questions?

Will this part hold on the Moon? Now you can ask in one sentence.

The team

Arav Gandhi, frontend, UX and backend

Braxton Anderson, frontend and backend

Amr Mohieldin and Jeremy Lin, backend assistants

Kevin Guo, pitch, presentation and team management

Built with TypeScript, React, three.js, Vite, Python, FastAPI, gmsh, Ansys MAPDL and the Gemini API